HOW TO CREATE AN ELASTIC BEANSTALK ON AWS

Technology

December 26, 2025 Admin Leave A Comment

WHAT IS AN ELASTIC BEANSTALK: An Elastic Beanstalk application is a container for Elastic Beanstalk components, including environments where your application code runs on platforms provided and managed by Elastic Beanstalk, or in custom containers that you provide.

THROUGH UNDERSTANDING OF ENVIROMENT ELASTIC BEANSTALK
An Elastic Beanstalk environment is a collection of AWS resources running together including an Amazon EC2 instance. When you create an environment, Elastic Beanstalk provisions the necessary resources into your AWS account.

STEP 1 – CREATE AN APPLICATION **
To create your example application, you'll use the Create application console wizard. It creates an Elastic Beanstalk application and launches an environment within it.
Reminder: an environment is a collection of AWS resources required to run your application code.

**STEP 2 – CONFIGURE THE ENVIROMENT
You choose the type of application you are deploying (a Web Server environment for handling web traffic), assign an application name (Deepdivedec22app) to organize all related deployments, and optionally add tags for easier management and billing, which together define the basic structure and identity of your Elastic Beanstalk application

STEP 3 – PROCEEDURE IN CONFIGURE ENVIROMENT
In this step, you assign a unique environment name (Deepdivedec22app-env), optionally choose a custom subdomain for your app's public URL, and select the application's runtime platform (Node.js 24 on Amazon Linux 2023 with version 6.7.1), which together determine how and where your application will run in the cloud

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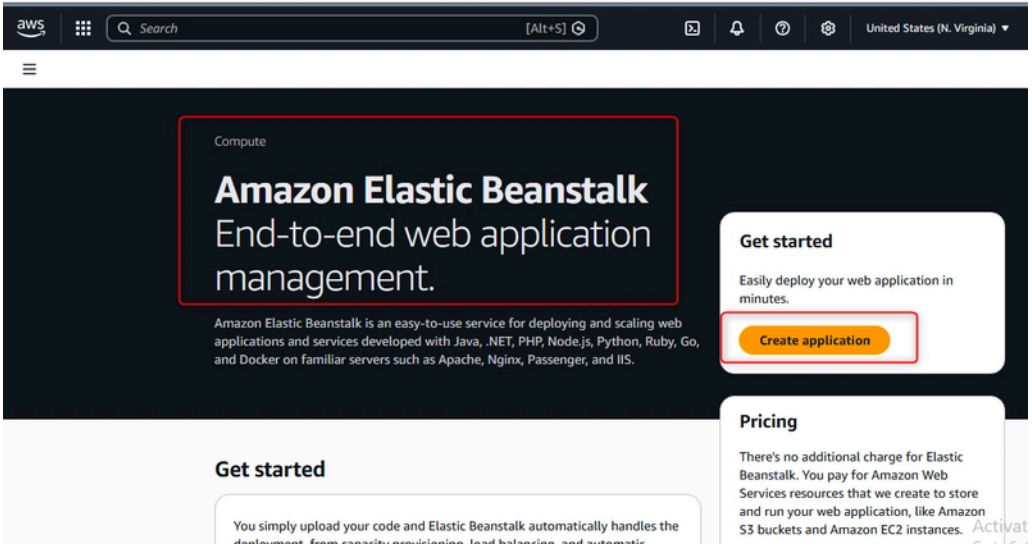
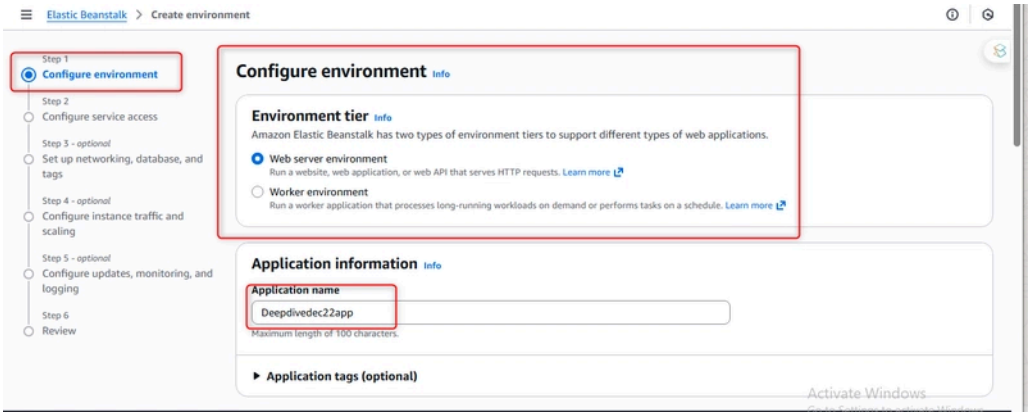
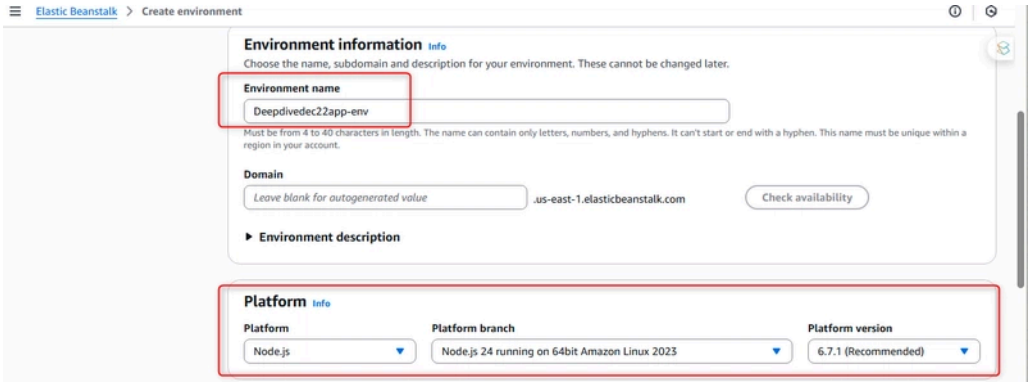
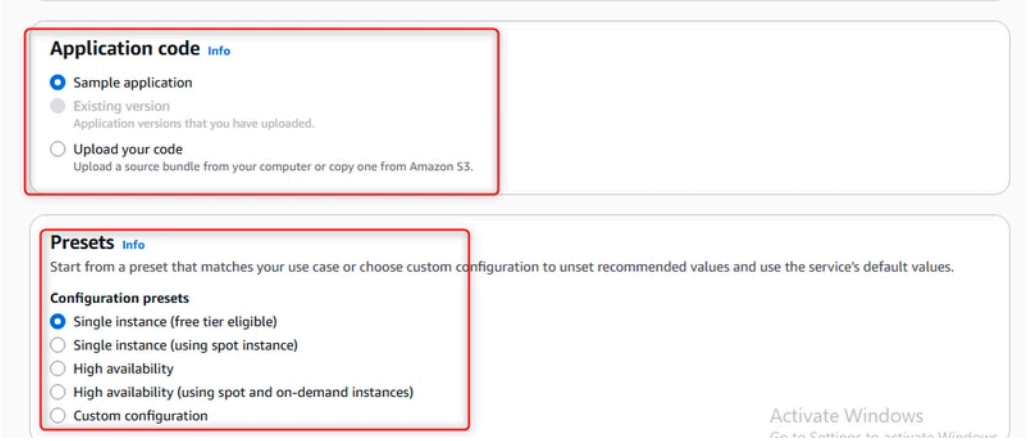




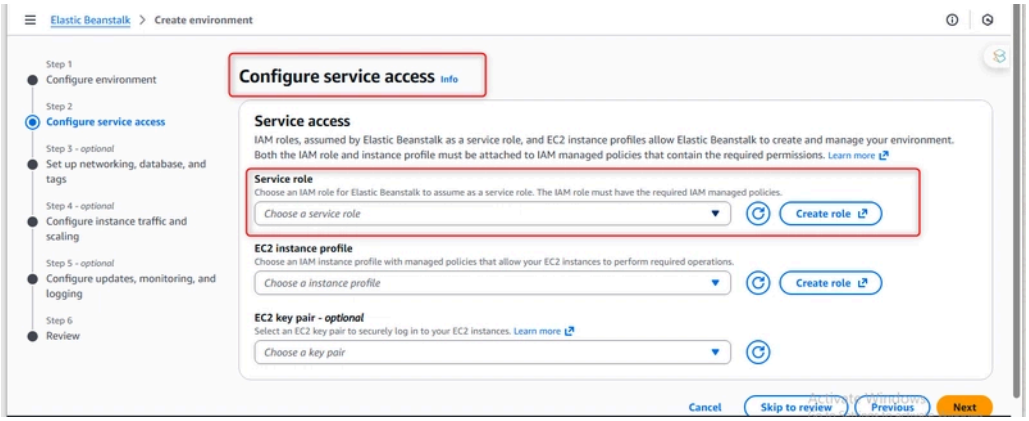


- STEP 4 NEXT PROCEDURE ON CONFIGURE ENVIROMENT *
  IT is a configuration step from a cloud service deployment interface (likely AWS Elastic Beanstalk) where users must upload their application's source code and then select a pre-defined infrastructure template, or "preset," which automatically sets parameters like server type, scaling, and availability to match common deployment scenarios such as a simple test setup or a high-availability production environment.



- STEP 5 CONFIGURE SERVICE ACCESS *

  A configuration step for setting up service access, which is critical for security and functionality in a managed cloud service like AWS Elastic Beanstalk. Here, you must define two essential Identity and Access Management (IAM) roles: a service role that grants the Elastic Beanstalk platform itself the permissions it needs to manage resources on your behalf, and an EC2 instance profile that provides the necessary permissions to the actual application servers (EC2 instances) it launches. Additionally, there is an optional setting to select an EC2 key pair, which is used for secure SSH access to those instances for administrative purposes.



STEP 5 HOW TO CONFIGURE SERVICE ROLE

This image Below shows the first step in creating an IAM role, where you must specify the trusted entity—the type of principal (like an AWS service, another account, or a user from an external identity provider) that is allowed to “assume” or use this role to perform actions in your AWS account. Selecting the correct trusted entity is the foundational security step that defines who or what can be granted the permissions attached to this role.

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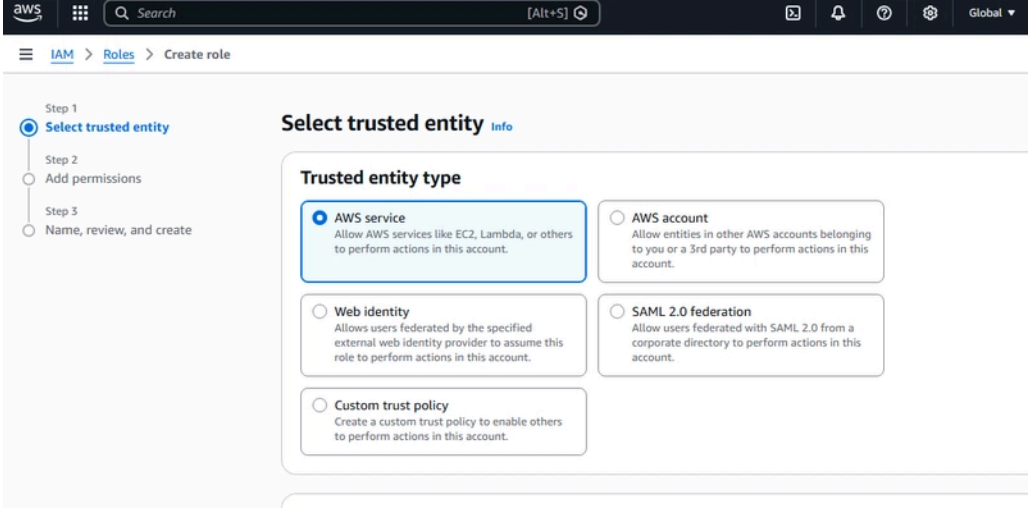
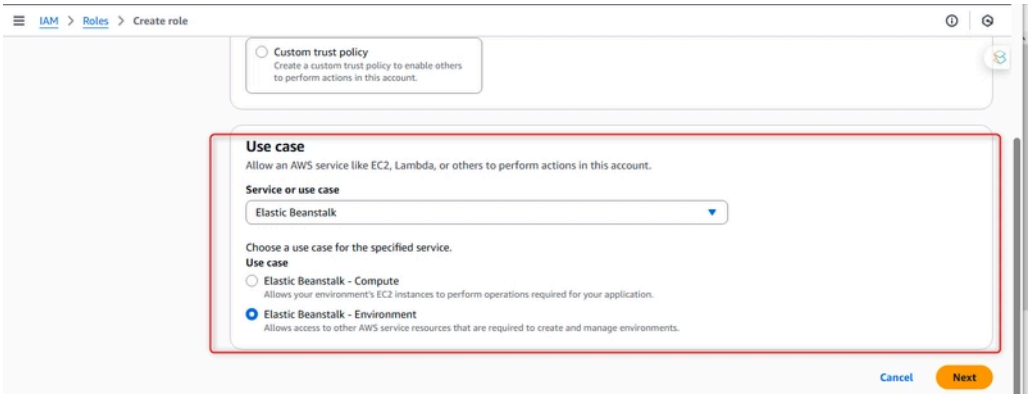


- STEP 6 ROLE ACCESS CONTINUATION *
  This image Below shows the continuation of creating an IAM role for AWS Elastic Beanstalk, focusing on selecting a specific use case for the service. After choosing “Elastic Beanstalk” as the service, you are presented with two distinct use cases that define the role’s purpose.

The “Elastic Beanstalk – Compute” use case is designed for an EC2 instance profile. This role is attached to the application servers themselves, granting them permissions to perform actions necessary for the application to run, such as uploading logs to Amazon S3 or sending metrics to Amazon CloudWatch.

The “Elastic Beanstalk – Environment” use case is designed for the service role. This role is assumed by the Elastic Beanstalk platform service, giving it permission to manage AWS resources like EC2 instances, Auto Scaling groups, and load balancers on your behalf to create and manage the environment’s infrastructure. Selecting the correct use case here is crucial for ensuring the right permissions are granted to the right component of the Elastic Beanstalk architecture.



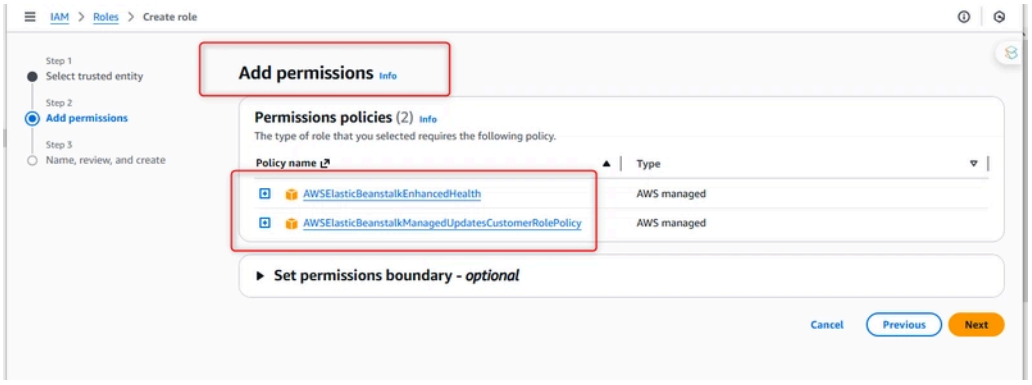
- STEP 7 ROLE ACCESS CONTINUATION *
  This image shows the next step in creating an IAM role for Elastic Beanstalk, where the necessary permissions policies are being attached.

When you select a specific use case (like “Elastic Beanstalk – Environment”), AWS automatically determines which managed policies are required for that role to function properly. In this case, the platform is pre-selecting two AWS-managed

policies: AWSElasticBeanstalkEnhancedHealth and AWSElasticBeanstalkManagedUpdatesCustomerRolePolicy.

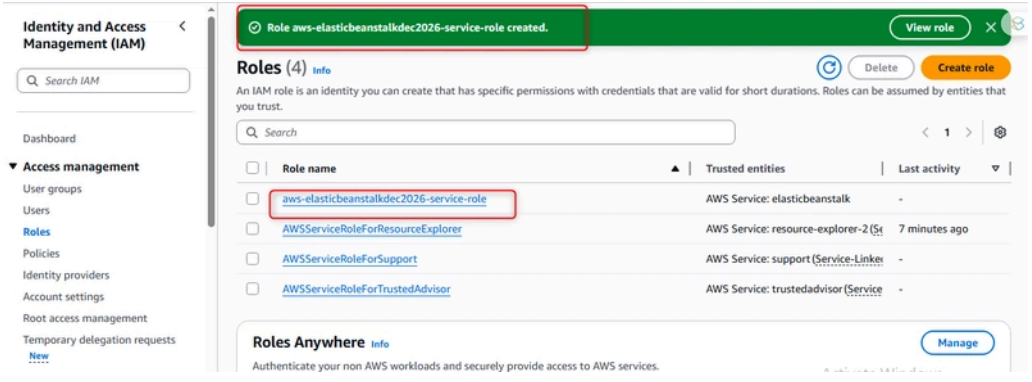
The first policy grants permissions for Elastic Beanstalk to collect and report enhanced health monitoring information from your environment. The second policy allows Elastic Beanstalk to perform managed updates on your environment, such as applying platform updates in a controlled way.

This step automates security best practices by attaching the minimum, predefined permissions needed for the role's intended function, as defined by AWS.



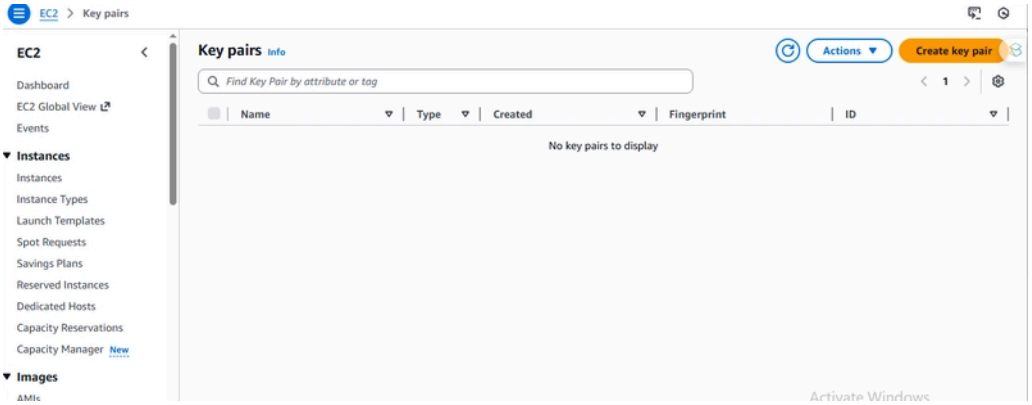
- STEP 8 CONTINUATION ROLE ACCESS ON SERVICE ROLE FINAL STEP * This image shows the final step in the IAM role creation process within the AWS Management Console, confirming the successful creation of the role. The newly created role, named aws-elasticbeanstalkdec2026-service-role, is now visible in the list of IAM roles.

The screenshot highlights key details about the role: it is listed under "Access management," and its trusted entity is confirmed to be the "elasticbeanstalk" AWS service, meaning only the Elastic Beanstalk platform can assume this role. The creation is marked as successful, and the role is ready to be selected in the previous "Configure service access" step, completing the setup of the required service role for deploying an environment.



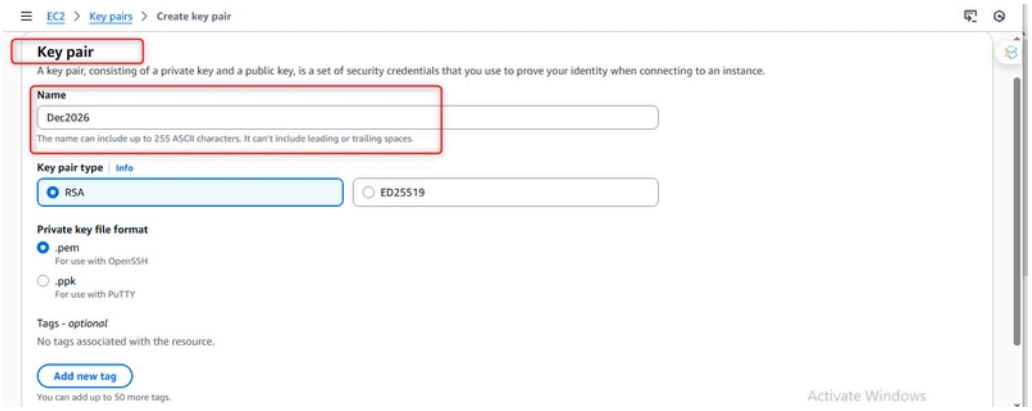
STEP 9 CREATING A PAIR KEY FOR SEVICE ACCESS
From the Amazon EC2 console, specifically the Key Pairs section. This is where you manage the SSH keys needed to securely log into your EC2 instances.

Right now, I'm looking at the list view where I can search for any existing key pairs by name, type, or other details. To make a new one, I would click the "Create key pair" button in the actions menu. You can see the rest of the EC2 navigation menu on the left, showing options for Instances, Images, and other resources, which tells me this key management is just one part of the overall process for setting up a server.



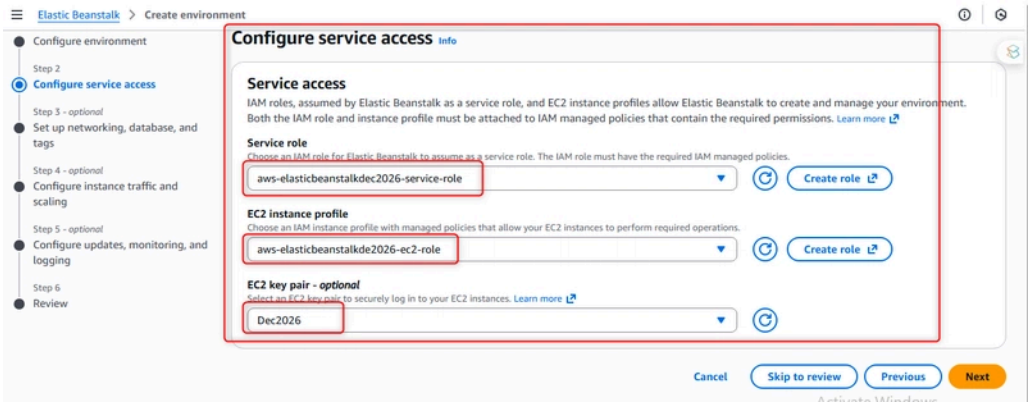
STEP 10 CONTINUATION CREATING A PAIR KEY FOR SEVICE ACCESS

Here, I need to define the key's properties. I've named it "Dec2026" and must select a Key pair type (choosing between the RSA or ED25519 encryption algorithms). I also need to choose the Private key file format—.pem for use with standard OpenSSH clients on Linux or Mac, or .ppk for use with the PuTTY client on Windows. There's also an optional section to add tags for better resource organization. Once I fill this out and create the key, the .pem or .ppk file will be automatically downloaded to my computer; this private key must be kept secure, as it's my only way to authenticate via SSH.

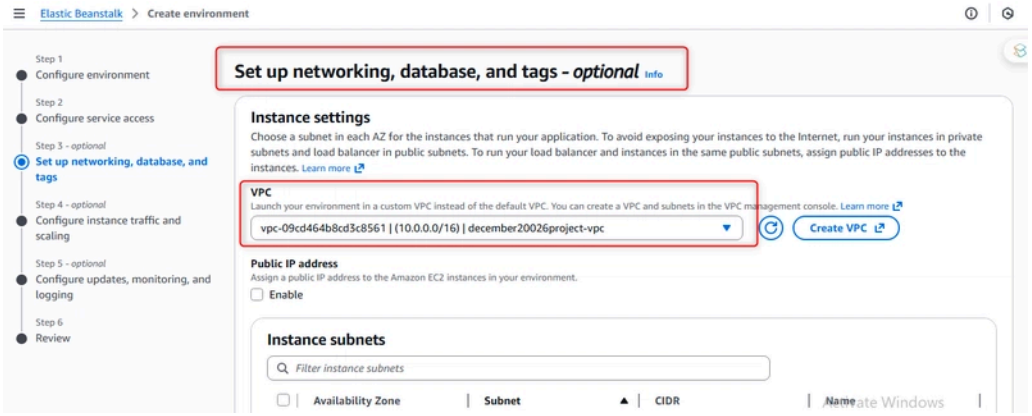


STEP 11 FINAL STEP FOR CONFIGURING SERVICE ACCESS

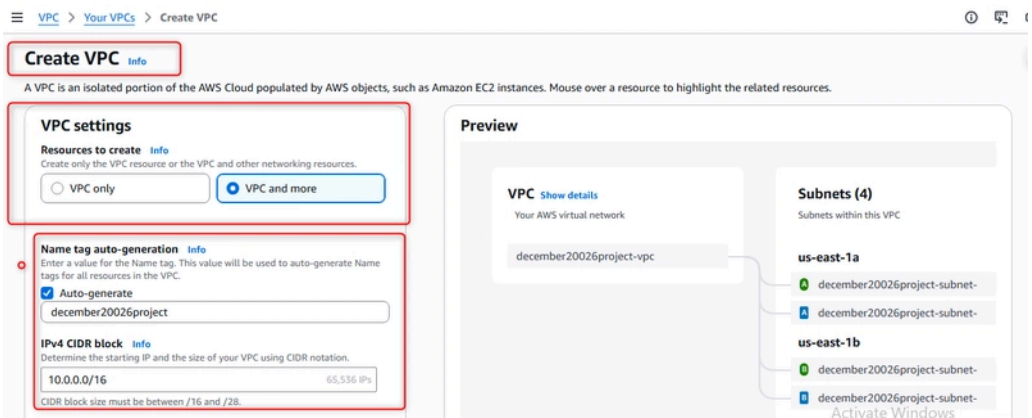
I have successfully configured all the necessary security identities. The previously created service role (aws-elasticbeanstalkdec2026-service-role) and EC2 instance profile (aws-elasticbeanstalkdec2026-ec2-role) are now selected, granting the platform and the application servers their required permissions. Additionally, I've chosen the optional EC2 key pair named "Dec2026," which will allow me to SSH into the instances for direct access if needed. With all these elements in place, I can proceed to the next optional steps for networking and scaling, or go straight to reviewing and launching the environment.



- STEP 12 HOW TO SET UP NETWORKING DATABASE AND TAG *
  Here, I'm configuring the network layout for my application. I've selected a custom VPC named december20026project-vpc instead of using the default one. The main decision on this screen is whether to assign a public IP address to the EC2 instances. Enabling this would allow the instances to be directly reachable from the internet, which is a key security and architectural choice. The interface also shows where I can choose specific subnets within the VPC to launch the instances into.

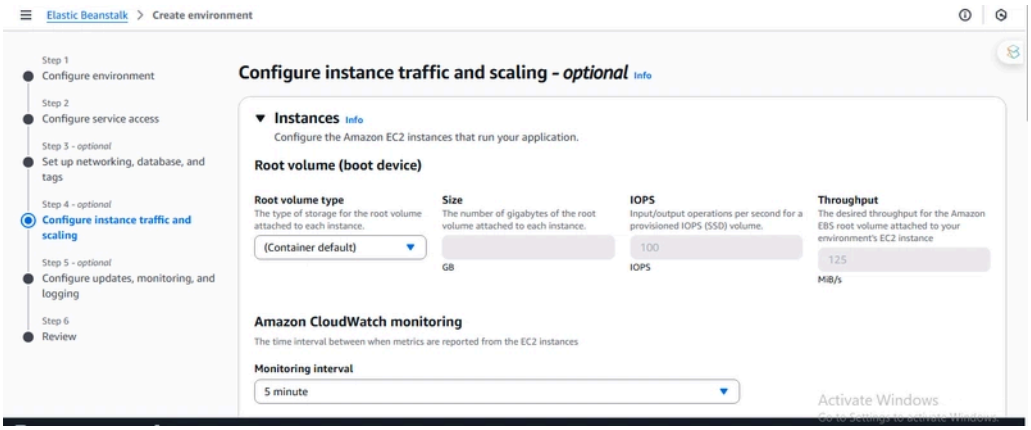


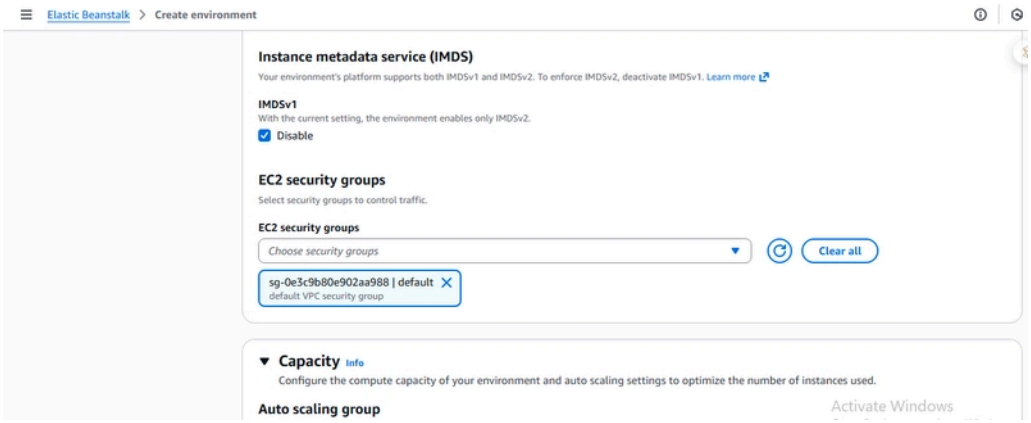
- STEP 13 CREATE A VPC *
  I'm choosing the option to create "VPC and more", which will automatically generate a complete, ready-to-use network including subnets and other core resources. I've set the Name tag to december20026project, and the VPC will be named december20026project-vpc. For the IP address range, I've specified the CIDR block 10.0.0.0/16, which provides over 65,000 private IP addresses. The preview shows that this setup will automatically create four subnets across two Availability Zones (us-east-1a and us-east-1b), establishing a foundational and redundant network structure for my application.

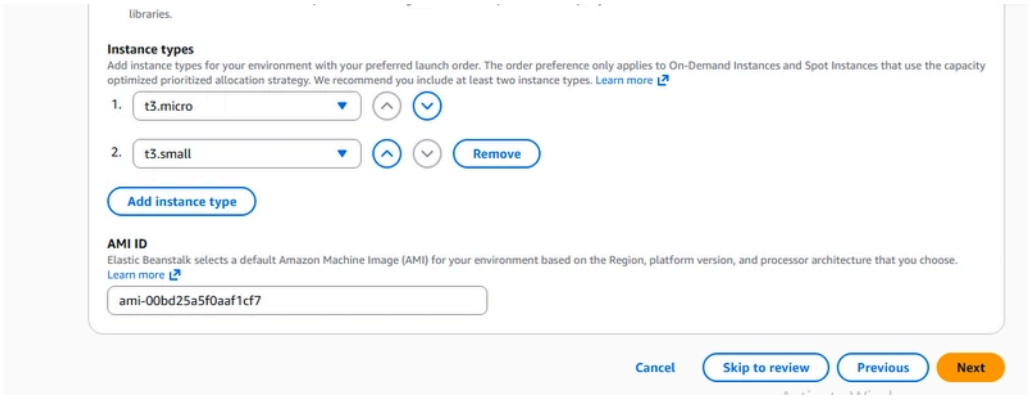


STEP 14 CONFIGURE INSTANCE TRAFFIC AND SCALING

Right now, I'm specifically looking at the settings for the root volume—which is basically the main hard drive for each server. Here, I can choose the type of storage, decide how big it should be in gigabytes, and set performance details like the IOPS (how many read/write operations it can handle per second) and throughput (the data transfer speed). At the bottom, I'm also setting up CloudWatch monitoring to report metrics from the servers every five minutes, so I can keep an eye on their performance.



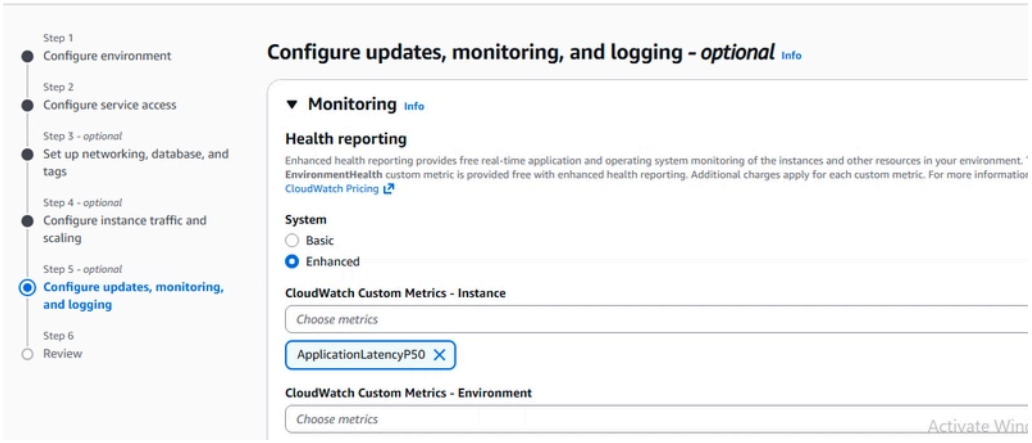


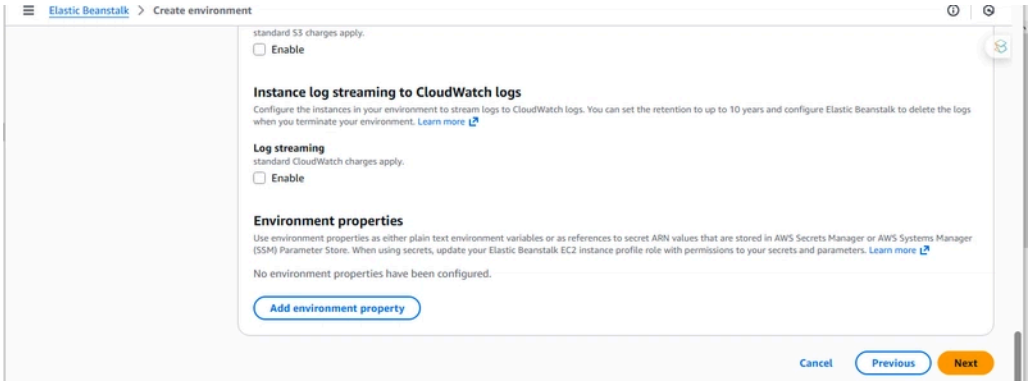


- STEP 15 CONFIGURE UPDATE MONITORING AND LOGGING *
  I've completed all the setup steps: from the initial configuration and security access to the optional networking and server settings. This review page gives me one last chance to check every decision I've made, from the application name and platform to the chosen instance type, VPC, key pair, and monitoring options.

Scrolling through this, I can see all my configured details listed in a clear summary, including the service role, instance profile, and that I've enabled enhanced health

reporting. Seeing everything confirmed here makes me confident I've set it up correctly. Once I click "Submit," Elastic Beanstalk will start provisioning all the AWS resources—like the load balancer, EC2 instances, and Auto Scaling group—to build and launch my environment.

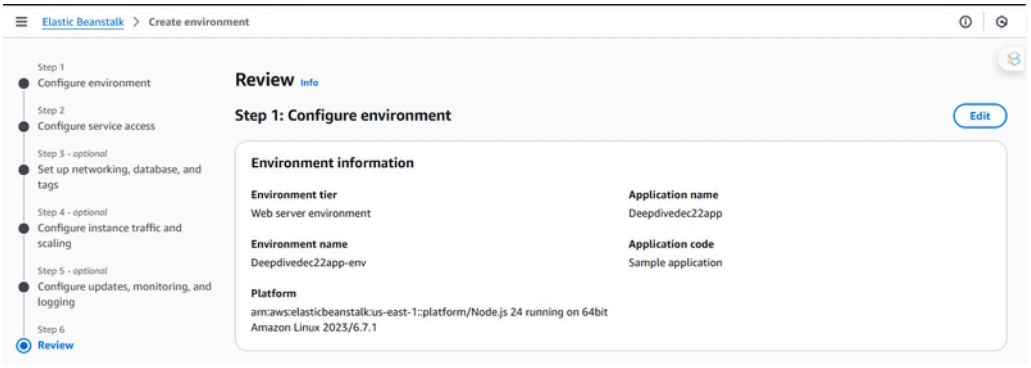


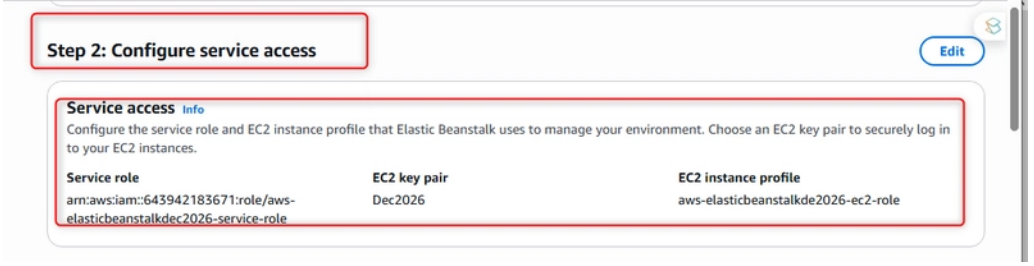


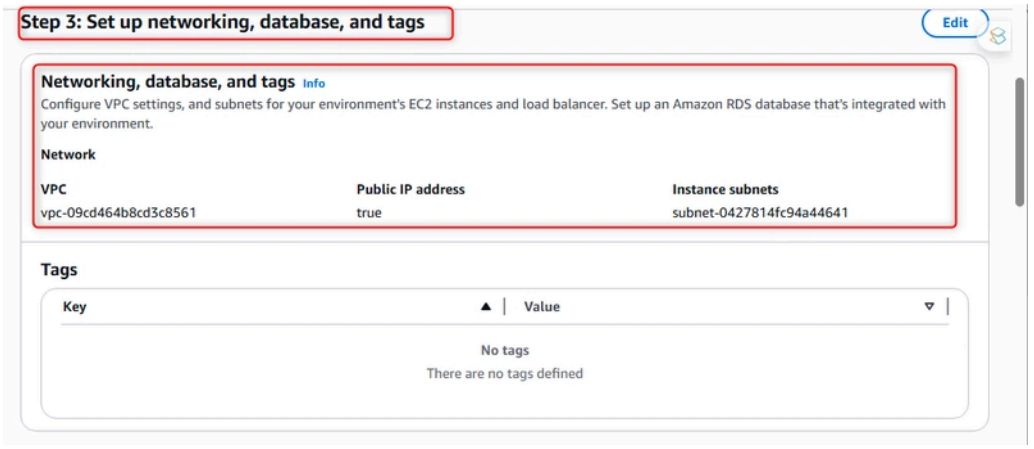
STEP 16 FINAL REVIEW BEFORE DEPLOYMENT OF APP
I'm on the final review page of the Elastic Beanstalk setup, where I'm double-checking all my configuration choices before launching the environment.

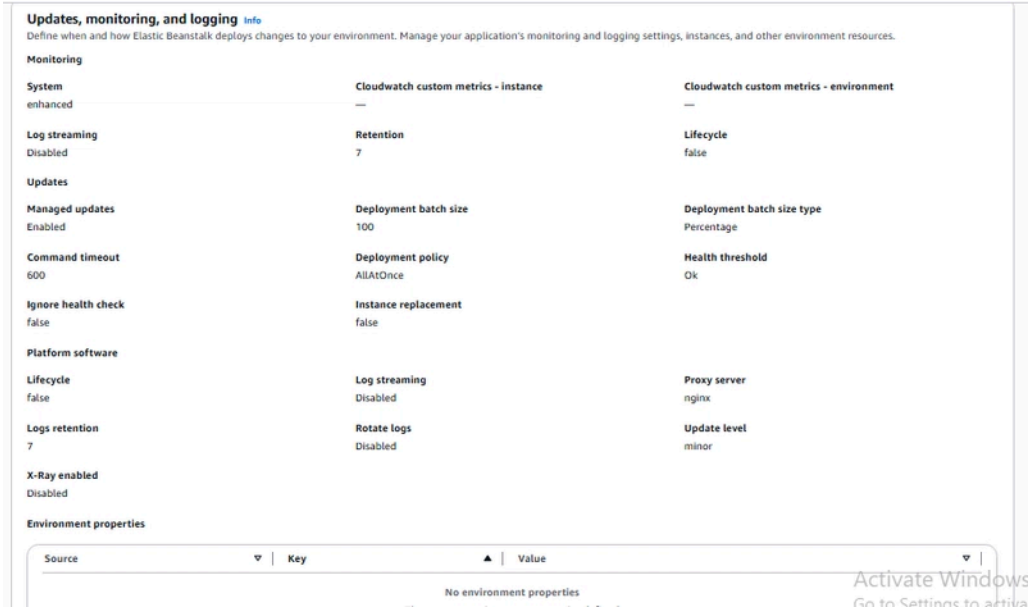
This first section of the review summarizes the core setup from Step 1. I can see that my environment, named Deepdivedec22app-env, is configured as a Web server environment. It's set to run on the Node.js 24 platform on Amazon Linux. The application itself is named Deepdivedec22app, and for this initial deployment, I'm using the sample application code provided by AWS.

The progress bar on the right confirms that I have already configured the service access, networking, instance settings, and monitoring in the previous steps. Everything looks correct here, so I'm ready to proceed to the final launch.

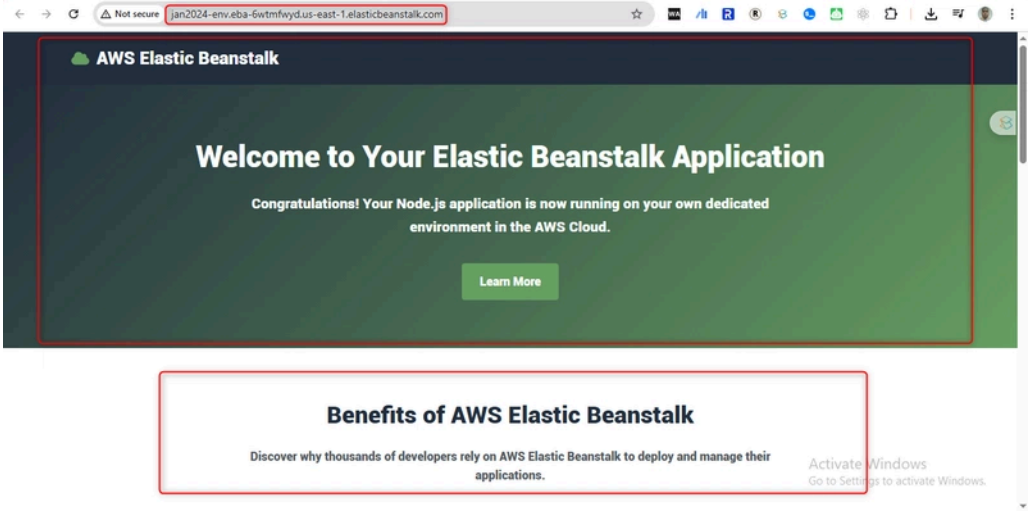








STEP 17 FINAL DEPLOYEMENT



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